

Notes: Cremers and Nair (JF, 2005)

Governance Mechanisms and Equity Prices

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Conceptual framework	3
3.1	External governance: takeover vulnerability	3
3.2	Internal governance: active shareholders	3
3.3	Substitutes versus complements	3
4	Data and measurement	4
4.1	Sample and time period	4
4.2	External governance variables	4
4.3	Internal governance variables	4
4.4	Returns and performance variables	4
5	Empirical strategy	5
5.1	Portfolio sorts	5
5.2	Two-step WLS regression	5
5.3	Robustness and interpretation tests	5
6	Main findings	6
6.1	External governance matters only when internal governance is high	6
6.2	Internal governance matters mainly when external governance is high	6
6.3	Size and leverage condition the effect	6
6.4	Alternative takeover index confirms the pattern	6
6.5	Profitability supports complementarity	6
6.6	Firm values and cash-flow risk complicate interpretation	7
7	Original figures and tables with reading notes	7
7.1	Table I: Distribution of Firms Based on Takeover Vulnerability and Active Shareholders and Table II: Correlation Matrix	7
7.2	Table III: Importance of External Governance Conditional on the Level of Internal Governance and Table IV: Importance of Internal Governance Conditional on the Level of External Governance	8
7.3	Table V: The Interaction between Governance Mechanisms and Firm Characteristics and Table VI: Summary Statistics: Alternative Measure for Takeover Defense (ATI)	9
7.4	Table VII: The Effect of Size and Leverage – Using ATI and Figure 1: Intertemporal Pattern with Public Pension Fund Ownership	10
7.5	Table VIII: Governance Mechanisms and Firm Profitability and Table IX: Governance Mechanisms and Firm Valuations	11
7.6	Table X: Governance Mechanisms and Cash Flow Risk	12

8	How to read the tables together	12
8.1	The return evidence	12
8.2	The interaction evidence	12
8.3	The fundamentals evidence	13
9	Mechanism intuition	13
9.1	Why might blockholders and takeovers complement each other?	13
9.2	Why might takeovers make blockholders more effective?	13
9.3	Why does leverage matter?	13
10	Limitations and cautions	13
11	Bottom line	14

1 Big picture

- **Question.** How do internal governance mechanisms, especially active shareholders, interact with external governance mechanisms, especially takeover vulnerability and the market for corporate control?
- **Core idea.** Governance mechanisms need not work independently. A blockholder or public pension fund may make takeovers more likely or more credible; the takeover market may make shareholder monitoring more valuable. The paper tests whether the two mechanisms are substitutes or complements.
- **Empirical setting.** The authors study U.S. public firms from 1990 to 2001. External governance is measured using antitakeover provisions from IRRC/Gompers-Ishii-Metrick style data; internal governance is measured using institutional blockholder ownership and public pension fund ownership.
- **Main result.** Internal and external governance mechanisms are complements. Portfolios based only on takeover vulnerability or only on active shareholder ownership are much weaker than portfolios that combine the two mechanisms.
- **Magnitude.** A portfolio that buys firms with high takeover vulnerability and high internal governance and shorts firms with low takeover vulnerability and high internal governance generates roughly 10–15% annualized abnormal returns, depending on the internal governance proxy and weighting scheme.
- **Mechanism and heterogeneity.** The complementarity effect is strongest among low-leverage firms and appears stronger for smaller firms. This is consistent with theories in which debt can reduce takeover probability and large size makes external control harder.
- **Interpretation.** The authors do not claim a clean causal design. They interpret the abnormal returns through competing lenses: mispricing of governance complementarity, an omitted risk factor related to governance, or an unobserved factor correlated with the governance portfolios.

2 Incremental contribution of the paper

- **Interaction rather than isolated governance.** Earlier work often asks whether one governance index predicts returns or firm value. This paper asks whether external takeover pressure and internal monitoring jointly determine outcomes.
- **Substitutes versus complements.** The paper provides an empirical test of two theoretical views. Under substitution, one strong governance mechanism should offset weakness in another. Under complementarity, one mechanism becomes valuable mainly when the other is also present.
- **Moves beyond turnover-based tests.** Prior studies often use CEO turnover to infer governance effectiveness. The paper instead studies equity returns, firm values, and accounting profitability, avoiding the selection problem that turnover only observes cases where the threat is exercised.
- **New external governance proxy.** In addition to the broad Gompers-Ishii-Metrick index, the authors build a parsimonious Alternative Takeover Index (ATI) based on three key takeover

barriers: preferred blank check, staggered boards, and restrictions on special meetings or written consent.

- **Two internal governance proxies.** The paper separately studies largest institutional blockholder ownership and ownership by large public pension funds. This matters because public pension funds are viewed as relatively active monitors, while blockholder ownership captures concentrated voting power more broadly.
- **Firm-characteristic interaction.** The paper explicitly tests whether size and leverage condition the complementarity effect, linking governance returns to theories of capital structure and corporate control.
- **Cross-validation with fundamentals.** The paper does not stop at stock returns. It also asks whether the same governance interaction appears in ROA, ROE, net profit margin, Q, and cash-flow risk.

3 Conceptual framework

3.1 External governance: takeover vulnerability

External governance is the disciplinary force of the market for corporate control. A firm that is vulnerable to takeovers faces a stronger threat that poorly performing managers can be replaced or that assets can be redeployed by an acquirer.

Interpretation: The paper does not equate all shareholder-rights variables with governance in general. It interprets the GIM index mainly as an antitakeover index. Higher takeover vulnerability means stronger external governance.

3.2 Internal governance: active shareholders

Internal governance is shareholder monitoring from investors with enough incentive or voting power to discipline management. The paper focuses on two proxies:

- **BLOCK:** percentage ownership by the largest institutional blockholder, where a blockholder has at least 5% of shares.
- **PP:** percentage ownership by a group of the 18 largest public pension funds.

Economic message: A large shareholder may internalize enough of the takeover gain to monitor management, coordinate with other shareholders, or reduce free-riding in takeover situations. Public pension funds are also less likely than some other institutions to be constrained by business ties with management.

3.3 Substitutes versus complements

Two empirical patterns are possible:

- **Substitution.** Strong internal governance and strong external governance are alternative ways to discipline management. If this is true, one mechanism should matter even when the other is weak, and having both should not be especially different.

- **Complementarity.** Internal governance makes external governance more effective, and external takeover pressure makes internal monitoring more valuable. If this is true, returns or performance should be strongest only when both mechanisms are active.

Takeaway: The paper finds the complementarity pattern: external governance matters mainly when active shareholders are present, and internal governance matters mainly when takeover vulnerability is high.

4 Data and measurement

4.1 Sample and time period

The equity-return tests cover September 1990 to December 2001. The sample begins with firms for which takeover-defense data are available from IRRC publications. Firms with dual-class common stock are excluded. The resulting sample averages roughly 1,500 firms per year.

4.2 External governance variables

The first external governance measure is based on the Gompers-Ishii-Metrick governance index G , which counts 24 provisions that restrict shareholder rights or protect management. The authors transform this into:

$$EXT = 24 - G.$$

A higher value of EXT means greater takeover vulnerability and therefore stronger external governance.

The second external governance measure is:

$$ATI = 3 - (\text{preferred blank check} + \text{staggered board} + \text{meeting/written-consent restrictions}).$$

A firm with $ATI = 3$ has none of these three key defenses and is most vulnerable to takeovers. A firm with $ATI = 0$ has all three and is least vulnerable.

4.3 Internal governance variables

The internal governance proxies come from institutional shareholdings:

- $BLOCK$ is the percentage of shares held by the largest institutional blockholder.
- $TOTBLOCK$ is the percentage of shares held by all institutional blockholders; this appears in descriptive statistics, but the main tests emphasize $BLOCK$.
- PP is the percentage held by the 18 largest public pension funds.

4.4 Returns and performance variables

Monthly stock returns come from CRSP. Abnormal returns are estimated using the Carhart four-factor model:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_M MKT_t + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t + \varepsilon_{p,t}.$$

The key outcome in the portfolio tests is the annualized alpha α_p .

For fundamentals, the paper studies industry-adjusted ROA, ROE, net profit margin, and firm Q . Leverage is adjusted by industry median using Fama-French industry classifications. The low-leverage subsample is the bottom third of industry-adjusted leverage.

5 Empirical strategy

5.1 Portfolio sorts

The paper sorts firms into governance portfolios using external and internal governance categories. The basic tests ask whether long-short portfolios based on one governance mechanism produce abnormal returns conditional on the level of the other mechanism.

- To test external governance conditional on internal governance, the paper buys high-takeover-vulnerability firms and shorts low-takeover-vulnerability firms within a given internal governance category.
- To test internal governance conditional on external governance, the paper buys high-blockholder or high-public-pension-fund firms and shorts low-internal-governance firms within a given external governance category.

Interpretation: If external governance is independently valuable, the external-governance spread should earn abnormal returns in both low and high internal-governance groups. If mechanisms are complements, the spread should appear only when internal governance is high.

5.2 Two-step WLS regression

The paper also estimates a two-step weighted least squares design. First, firms are sorted into portfolios by external governance, internal governance, size, and leverage; portfolio alphas are estimated using the Carhart model. Second, those alphas are regressed on functions of governance categories.

The regression uses two summary variables:

$$Max = \max(\text{external governance level}, \text{internal governance level}),$$

$$Min = \min(\text{external governance level}, \text{internal governance level}).$$

Economic message: Max captures substitution: if either mechanism is strong, the maximum should explain returns. Min captures complementarity: the minimum is high only when both mechanisms are high. A positive coefficient on Min is the direct complementarity evidence.

5.3 Robustness and interpretation tests

The paper conducts several robustness exercises:

- Replacing EXT with ATI .
- Studying the intertemporal pattern of 36-month moving average alphas.
- Removing target firms and bidder firms to test whether acquisition-announcement effects mechanically drive returns.
- Testing whether accounting profitability also displays the same complementarity.

- Studying firm Q and cash-flow volatility to assess whether the abnormal returns might be compensation for risk rather than mispricing.

6 Main findings

6.1 External governance matters only when internal governance is high

Table III is the central return result. When internal governance is high, a high-minus-low takeover vulnerability portfolio earns large abnormal returns. With *BLOCK* as the internal governance proxy, the high-internal-governance portfolio produces a value-weighted alpha of 10.83% and an equal-weighted alpha of 14.98%. With *PP*, the corresponding value-weighted and equal-weighted alphas are 9.46% and 10.84%.

Takeaway: The broad external governance effect documented in earlier studies is concentrated in firms with active internal monitoring.

6.2 Internal governance matters mainly when external governance is high

Table IV reverses the conditioning. When external governance is weak, internal governance produces little or no abnormal return. When takeover vulnerability is high, the internal governance spread becomes economically meaningful. The result is strongest for *BLOCK*.

Interpretation: This is the second half of the complementarity argument. Active shareholders are not enough by themselves; their monitoring appears more valuable when the takeover market provides a credible external threat.

6.3 Size and leverage condition the effect

Table V shows that complementarity is stronger in small firms and low-leverage firms. The *Min* variables are positive and significant, especially with low leverage.

Economic message: Low leverage may make firms more contestable because debt can reduce takeover probability or affect takeover incentives. Smaller firms may be easier to acquire, making external governance more effective.

6.4 Alternative takeover index confirms the pattern

Tables VI and VII repeat the analysis using *ATI*, the narrower index of takeover vulnerability. Despite fewer firms in the strongest external governance category, the complementarity evidence remains.

Takeaway: The main result is not mechanically driven by the broad 24-provision GIM index. It survives when external governance is measured using only three key takeover-defense provisions.

6.5 Profitability supports complementarity

Table VIII shows that in low-leverage firms, high takeover vulnerability is associated with higher profitability only when internal governance is high. When internal governance is low, the profitability differences are small or absent.

Interpretation: This helps rule out a purely price-based story. The return patterns line up with accounting outcomes, suggesting that governance complementarity is related to real performance, not only to market sentiment.

6.6 Firm values and cash-flow risk complicate interpretation

Table IX shows that higher *EXT* or *ATI* and higher internal governance are individually associated with higher *Q*, but the interaction terms are negative. This means firms with both mechanisms active are not valued as highly as a simple additive interpretation would predict.

Table X shows that the high-high governance group often has higher cash-flow volatility, especially in net profit margins. The authors interpret this as evidence consistent with an omitted-risk interpretation.

Caveat: The paper is careful: it cannot fully distinguish mispricing from risk. The abnormal returns may reflect investors initially failing to price governance complementarity, or they may reflect a priced risk factor omitted from the Carhart model.

7 Original figures and tables with reading notes

7.1 Table I: Distribution of Firms Based on Takeover Vulnerability and Active Shareholders and Table II: Correlation Matrix

Table I Distribution of Firms Based on Takeover Vulnerability and Active Shareholders				
For each of the four dates for which the IRRC data are updated, this table reports some summary statistics of our sample of firms. The variable <i>EXT</i> denotes the level of takeover vulnerability for a firm, with higher values indicating greater vulnerability to takeovers, and is equal to 24-G, where G is the proxy created by Gompers et al. (2003) using the IRRC publications. The top panel reports the number of firms that are classified as having low and high <i>EXT</i> , using two different cutoff levels for these classifications. Further, we report the 25%, 50%, and the 75% percentiles of the proportion of the total shares outstanding of our sample of firms that is held by the largest institutional blockholder (<i>BLOCK</i> , middle panel) and by the group of the 18 largest public pension funds (<i>PP</i> , lower panel). Blockholders are defined as investors holding at least 5% of outstanding shares. See the appendix for the list of public pension funds.				
	Sep-90	Jul-93	Jul-95	Feb-98
Number of Firms <i>EXT</i> $\geq$ 19	148	131	112	195
Number of firms <i>EXT</i> $\geq$ 18	262	212	215	346
Number of Firms <i>EXT</i> $\leq$ 10	83	89	84	77
Number of firms <i>EXT</i> $\leq$ 11	167	186	187	171
25% percentile of <i>BLOCK</i>	0%	0%	0%	0%
50% percentile of <i>BLOCK</i>	6.21%	6.92%	7.45%	8.11%
75% percentile of <i>BLOCK</i>	9.32%	9.91%	10.26%	11.22%
25% percentile of <i>PP</i>	0.66%	1.30%	1.58%	1.28%
50% percentile of <i>PP</i>	2.40%	2.88%	2.70%	2.16%
75% percentile of <i>PP</i>	4.25%	4.61%	3.86%	2.87%

Table II Correlation Matrix						
The matrix below gives the pairwise correlations between quartile-groups of <i>EXT</i> , <i>ATI</i> , <i>SIZE</i> , <i>BLOCK</i> , <i>TOTBLOCK</i> , and <i>PP</i> . Here, <i>EXT</i> denotes the firm-specific level of takeover vulnerability based on the index developed by Gompers et al. (2003) that incorporates 24 takeover provisions; <i>ATI</i> is an alternative takeover vulnerability proposed in this paper that incorporates only three takeover provisions; <i>SIZE</i> is given by each firm's total market capitalization of outstanding stocks; <i>BLOCK</i> is the percentage of stocks held by the largest institutional blockholder (at least 5%); <i>TOTBLOCK</i> is the percentage of stocks held by all institutional blockholders; <i>PP</i> is the percentage of stocks held by the group of public pension funds (see the Appendix for the list of public pension funds); and finally, <i>LEVERAGE</i> is the industry median-adjusted leverage. The Fama-French 48-industry classifications are used for the adjustment. The correlations are those of the quartile dummies.						
	<i>EXT</i>	<i>ATI</i>	<i>SIZE</i>	<i>BLOCK</i>	<i>TOTBLOCK</i>	<i>PP</i>
<i>ATI</i>	64%					
<i>SIZE</i>	-4%	-3%				
<i>BLOCK</i>	-3.4%	-1.9%	-9%			
<i>TOTBLOCK</i>	-4%	-2%	-12%	88%		
<i>PP</i>	-16%	-12%	26%	6%	6%	
<i>LEVERAGE</i>	-0.2%	-1.2%	43%	1.2%	1.9%	-0.2%

Table II: Correlation Matrix

Table I: Distribution of Firms Based on Takeover Vulnerability and Active Shareholders

Read: Table I establishes how many firms fall into the extreme takeover-vulnerability groups and reports the distribution of *BLOCK* and *PP*. Table II reports correlations among governance proxies, size, and leverage.

Economic message: Two facts matter for later tests. First, public pension fund ownership is positively related to size, while blockholder ownership is negatively related to size. Second, *EXT* and *ATI* are highly correlated, supporting the use of *ATI* as a narrower robustness proxy.

7.2 Table III: Importance of External Governance Conditional on the Level of Internal Governance and Table IV: Importance of Internal Governance Conditional on the Level of External Governance

Table III Importance of External Governance (Takeover Vulnerability) Conditional on the Level of Internal Governance							
We report the annualized mean, the annualized abnormal return (alpha), and its t -statistic of a (value-weighted, VW, and equal-weighted, EW) portfolio that buys firms in the highest category and shorts firms in the lowest category of external governance as measured by EXT, for a given category of internal governance (1, 2, 3, or 4) or independent of internal governance (All). We consider two different proxies for internal governance: BLOCK (Panel A) and PP (Panel B). The variable EXT denotes the firm-specific level of takeover vulnerability based on the index developed by Gompers et al. (2003) that incorporates 24 takeover provisions; BLOCK is the percentage of stocks held by the largest institutional blockholder (at least 5%); and PP is the percentage of stocks held by the group of public pension funds (see the Appendix for the list of public pension funds). We also consider a portfolio that buys firms with both high takeover vulnerability and high internal governance and shorts firms with both low takeover vulnerability and low internal governance (HH-LL). External governance is measured using firm-specific takeover provisions in place. The time period used is 1990 to 2001, and the alphas are relative to the four-factor Carhart (1997) model.							
Panel A: Using BLOCK as Proxy for Internal Governance							
VW Long-Short Portfolios				EW Long-Short Portfolios			
Mean	Alpha	t -stat	BLOCK	Mean	Alpha	t -stat	BLOCK
1.51%	2.64%	1.24	All	4.55%	7.08%	3.47	All
-0.24%	-0.51%	-0.15	1	4.90%	5.55%	1.55	1
3.09%	5.67%	1.4	2	0.73%	2.97%	0.8	2
2.36%	2.46%	0.54	3	5.59%	7.35%	2.36	3
8.94%	10.83%	3.13	4	8.50%	14.98%	4.8	4
9.80%	7.42%	1.60	HH-LL	10.46%	13.02%	4.01	
Panel B: Using PP as Proxy for Internal Governance							
VW Long-Short Portfolios				EW Long-Short Portfolios			
Mean	Alpha	t -stat	PP	Mean	Alpha	t -stat	PP
1.51%	2.64%	1.24	All	4.55%	7.08%	3.47	All
0.37%	1.94%	0.34	1	5.40%	10.18%	2.09	1
-1.84%	-0.95%	-0.24	2	4.16%	4.02%	1.19	2
2.93%	3.43%	0.96	3	3.25%	6.17%	2.39	3
6.79%	9.46%	2.11	4	7.26%	10.84%	3.08	4
2.51%	3.99%	0.82	HH-LL	4.87%	12.42%	2.92	

Table III: Importance of External Governance Conditional on the Level of Internal Governance

Table IV Importance of Internal Governance (Active Shareholders) Conditional on the Level of External Governance							
We report the annualized mean, the annualized abnormal return (alpha), and its t -statistic of a (value-weighted, VW, and equal-weighted, EW) portfolio that buys firms in the highest category and shorts firms in the lowest category of internal governance as measured by BLOCK (Panel A) and by PP (Panel B), for a given category of external governance as measured by EXT (1, 2, 3, or 4) or independent of external governance (All). The variable EXT denotes the firm-specific level of takeover vulnerability based on the index developed by Gompers et al. (2003) that incorporates 24 takeover provisions; BLOCK is the percentage of stocks held by the largest institutional blockholder (at least 5%); and, PP is the percentage of stocks held by the group of public pension funds (see the Appendix for the list of public pension funds). External governance is measured using firm-specific takeover provisions in place. The time period used is 1990 to 2001, and the alphas are relative to the four-factor Carhart (1997) model.							
VW Long-Short Portfolios				EW Long-Short Portfolios			
Mean	Alpha	t -stat	EXT	Mean	Alpha	t -stat	EXT
Panel A: Using BLOCK							
3.20%	-1.02%	-0.47	All	3.04%	1.85%	1.00	All
0.9%	-3.41%	-0.74	1	1.97%	-1.95%	-0.62	1
0.54%	-3.94%	-1.1	2	3.09%	1.9%	0.79	2
3.09%	-2.53%	-0.82	3	5.09%	2.21%	0.86	3
10.08%	7.94%	2	4	5.57%	7.48%	2.04	4
Panel B: Using PP							
1.10%	0.51%	0.19	All	-1.30%	-0.53%	-0.21%	All
-4.28%	-5.47%	-1.05	1	-2.39%	1.59%	0.44	1
-1.28%	-0.93%	-0.25	2	-3.3%	-1.39%	-0.52	2
1.75%	-0.2%	-0.04	3	2.26%	0.07%	0.02	3
2.14%	2.06%	0.42	4	-0.52%	2.24%	0.45	4

Table IV: Importance of Internal Governance Conditional on the Level of External Governance

Read: Table III conditions the takeover-vulnerability long-short portfolio on internal governance. Table IV conditions the internal-governance long-short portfolio on takeover vulnerability.

Interpretation: These two tables are the cleanest portfolio evidence for complementarity. Table III says takeover vulnerability matters mainly when active shareholders are present. Table IV says active shareholders matter mainly when takeover vulnerability is high.

7.3 Table V: The Interaction between Governance Mechanisms and Firm Characteristics and Table VI: Summary Statistics: Alternative Measure for Takeover Defense (ATI)

Table V The Interaction between Governance Mechanisms and Firm Characteristics		
This table presents the regression results for the alphas of 81 equal-weighted portfolios created from independent $3 \times 3 \times 3$ sorts on EXT, PP/BLOCK, SIZE, and LEVERAGE (see Table II for a description of these variables). Panel A gives the results using PP, and Panel B using BLOCK. We first estimate portfolio alphas and their variance-covariance matrix using the Carhart (1997) four-factor model. We then estimate a weighted least squares regression of these alphas on Max and Min times size dummies (I) and on Min times both size and leverage dummies (II), with the variance-covariance matrix of the alphas as the weighting matrix. The variable Max denotes the maximum of the level of EXT and PP/BLOCK, and captures the substitution effect, Min denotes the minimum of the level of EXT and PP/BLOCK, and captures the complementarity effect. Here, the level of EXT, PP or BLOCK equals 1, 2, or 3, based on the $3 \times 3 \times 3$ sorting of firms into portfolios. The variables Small and Large denote the dummies for the smallest third and the largest two-thirds of firms, respectively, and LowLev and HighLev denote dummies for the firms with the lowest third and highest third LEVERAGE, respectively. We present the regression coefficients and their t-statistics in parentheses.		
	I	II
Panel A: Using PP		
Max * Small	-0.13 (-0.11)	
Min * Small	3.15 (2.35)	
Min * Small * LowLev		4.30 (3.59)
Min * Small * HighLev		2.29 (1.91)
Max * Large	0.10 (0.12)	
Min * Large	2.31 (2.79)	
Min * Large * LowLev		3.61 (3.94)
Min * Large * HighLev		0.42 (0.46)
Panel B: Using BLOCK		
Max * Small	-0.27 (-0.26)	
Min * Small	5.26 (4.39)	
Min * Small * LowLev		3.52 (3.09)
Min * Small * HighLev		1.82 (1.59)
Max * Large	0.69 (0.88)	
Min * Large	2.43 (2.92)	
Min * Large * LowLev		3.70 (4.24)
Min * Large * HighLev		0.93 (1.07)

Table V: The Interaction between Governance Mechanisms and Firm Characteristics

Table VI Summary Statistics: Alternative Measure for Takeover Defense (ATI)				
This table reports the number of firms for each of the four possible values of the alternative takeover vulnerability index, ATI, which incorporates three takeover provisions (blank check preferred, staggered boards, and restrictions on calling special meetings or acting by written consent). Firms with ATI = 3 are classified as having the highest external governance and are most vulnerable for takeovers, and firms with ATI = 0 are classified as being least vulnerable. The four dates are those for which the IRRC database containing firm-specific takeover provisions are updated.				
	Number of Firms			
	Sep-90	Jul-93	Jul-95	Feb-98
ATI = 0	328	378	412	482
ATI = 1	410	427	459	563
ATI = 2	446	381	344	428
ATI = 3	145	129	99	92

Table VI: Summary Statistics: Alternative Measure for Takeover Defense (ATI)

Read: Table V implements the two-step WLS test using *Max* and *Min* of governance categories, interacted with size and leverage. Table VI describes the distribution of the narrower *ATI* proxy.

Economic message: The *Min* coefficients capture complementarity because *Min* is large only when both governance mechanisms are strong. The evidence is strongest in low-leverage groups, consistent with the idea that leverage can weaken takeover discipline.

7.4 Table VII: The Effect of Size and Leverage – Using ATI and Figure 1: Intertemporal Pattern with Public Pension Fund Ownership

Table VII The Effect of Size and Leverage—Using ATI		
This table presents the regression results for the alphas of 81 equal-weighted portfolios created from independent $3 \times 3 \times 3$ sorts on ATI, PP/BLOCK, SIZE, and LEVERAGE (see Table II for a description of these variables). Panel A gives the results using PP, and Panel B using BLOCK. We first estimate portfolio alphas and their variance-covariance matrix using the Carhart (1997) four-factor model. We then estimate a weighted least squares regression of these alphas on Max and Min times size dummies (I) and on Min times both size and leverage dummies (II), with the variance-covariance matrix of the alphas as the weighting matrix. The variable Max denotes the maximum of the level of ATI and PP/BLOCK, and captures the substitution effect; Min denotes the minimum of the level of ATI and PP/BLOCK, and captures the complementarity effect. Here, the level of ATI, PP, or BLOCK equals 1, 2, or 3, based on the $3 \times 3 \times 3$ sorting of firms into portfolios. The variables Small and Large denote the dummies for the smallest third and the largest two-thirds of firms, respectively, and LowLev and HighLev denote dummies for the firms with the lowest third and highest third LEVERAGE, respectively. We present the regression coefficients and their <i>t</i> -statistics in parentheses.		
	I	II
Panel A: Using PP		
Max * Small	-0.10 (-0.09)	
Min * Small	3.29 (2.20)	
Min * Small * LowLev		1.56 (1.18)
Min * Small * HighLev		1.30 (0.99)
Max * Large	-0.34 (-0.37)	
Min * Large	2.62 (2.45)	
Min * Large * LowLev		2.68 (2.65)
Min * Large * HighLev		0.54 (0.53)
Panel B: Using BLOCK		
Max * Small	0.50 (0.38)	
Min * Small	2.50 (1.78)	
Min * Small * LowLev		1.57 (1.16)
Min * Small * HighLev		0.37 (0.27)
Max * Large	0.66 (0.87)	
Min * Large	1.84 (1.77)	
Min * Large * LowLev		2.55 (2.46)
Min * Large * HighLev		-0.35 (-0.34)

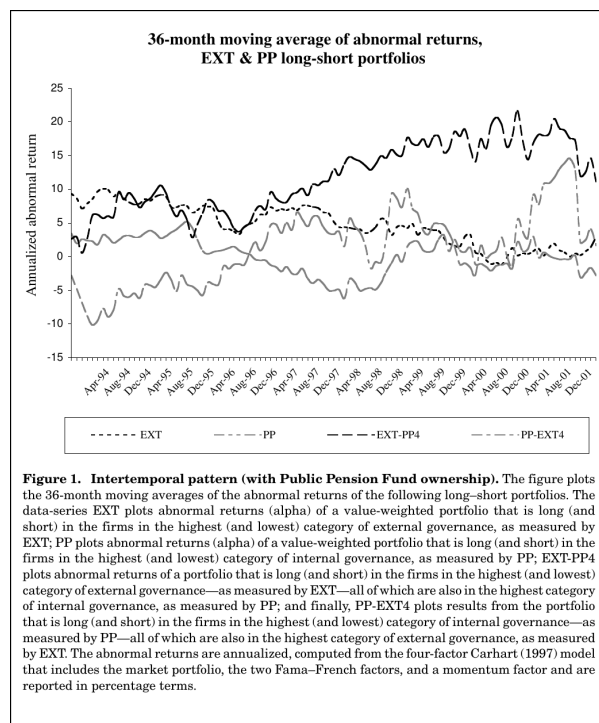


Figure 1: Intertemporal Pattern with Public Pension Fund Ownership

Table VII: The Effect of Size and Leverage – Using ATI

Read: Table VII repeats the size/leverage WLS analysis with *ATI*. Figure 1 plots 36-month moving averages of abnormal returns for governance spread portfolios based on *EXT* and *PP*.

Takeaway: The complementarity pattern is not a one-year artifact. Figure 1 shows that the combined external-internal portfolio is more consistently positive over time than either mechanism alone.

7.5 Table VIII: Governance Mechanisms and Firm Profitability and Table IX: Governance Mechanisms and Firm Valuations

Table VIII Governance Mechanisms and Firm Profitability			
The table reports the industry-adjusted average profitability of four different governance-level groups using the accounting measures return on assets (ROA), return on equity (ROE), and net profit margin (NPM), each in percentage points. We only use firms with the lowest third of LEVERAGE. Further, high and low refer to above and below the median, respectively, for the relevant governance variable. Panel A compares firms that have a high level of internal governance as measured by either PP or BLOCK but differ in their levels of external governance, as measured by EXT (see Table II for a description of these variables). Panel B compares firms that have a low level of internal governance but differ in their levels of external governance. Net profit margin is the ratio of income before extraordinary items available for common equity to sales; return on equity is the ratio of income before extraordinary items available for common equity to the sum of the book value of common equity and deferred taxes; return on assets is the ratio of income before extraordinary items to total assets. Each variable is adjusted for the industry median using the Fama and French (1997) industry classifications. The average for each group is reported, removing outliers (bottom and top 2.5%). The <i>t</i> -statistic for the differences in means test is also reported.			
EXT-INT	ROA	ROE	NPM
Panel A: Different Takeover Vulnerabilities When Internal Governance Is HIGH			
Using PP			
HIGH-HIGH	21.95	7.57	8.01
LOW-HIGH	16.55	5.92	5.16
<i>t</i> -statistic			
Difference in Means	(5.91)	(2.69)	(4.09)
Using BLOCK			
HIGH-HIGH	18.54	5.47	6.93
LOW-HIGH	13.71	3.38	3.28
<i>t</i> -statistic			
Difference in Means	(5.90)	(3.54)	(5.72)
Panel B: Different Takeover Vulnerabilities When Internal Governance Is LOW			
Using PP			
HIGH-LOW	16.55	4.15	4.99
LOW-LOW	16.32	4.65	5.64
<i>t</i> -statistic			
Difference in Means	(0.25)	(−0.72)	(−0.81)
Using BLOCK			
HIGH-LOW	19.85	6.29	5.94
LOW-LOW	19.49	7.69	7.82
<i>t</i> -statistic			
Difference in Means	(0.35)	(−1.99)	(−2.21)

Table VIII: Governance Mechanisms and Firm Profitability

Table IX Governance Mechanisms and Firm Valuations					
The table reports the average of the 11 cross-sectional estimates (Fama-MacBeth estimates) for regressions that use firm <i>Q</i> as the dependent variable. We include as controls the log of assets in the current fiscal year, firm age, and a dummy variable for inclusion in the S&P 500 as of the end of the previous year. The coefficients on the controls and the constant are omitted from the table. See Table II for a description of EXT and ATI, the proxies for takeover vulnerability, and BLOCK and PP, the proxies for internal governance. The variable <i>Q</i> is the ratio of the market value of assets to the book value of assets. The market value is calculated as the sum of the book value of assets and the market value of common stock less the book value of common stock and deferred taxes. The market value of equity is measured at the end of the current calendar year, and the accounting variables are measured in the current fiscal year. Industry adjustments are made by subtracting the industry median, where medians are calculated by matching the four-digit SIC codes from December of each year to the 48 industries designated by Fama and French (1997). The time-series averages (in percentage) and time-series <i>t</i> -statistics of the coefficients are reported in the table.					
Panel A: Using EXT as Proxy for Takeover Vulnerability					
	EXT	BLOCK	BLOCK × EXT	PP	PP × EXT
I.	3.93 (7.28)	1.05 (2.82)	−0.09 (3.45)		
II.	5.67 (6.00)			4.33 (2.60)	−0.78 (−3.86)
Panel B: Using ATI as Proxy for Takeover Vulnerability					
	ATI	BLOCK	BLOCK × ATI	PP	PP × ATI
I.	9.71 (5.47)	0.47 (2.29)	−0.21 (−3.31)		
II.	12.20 (5.91)			−1.51 (1.34)	−1.29 (−2.50)

Table IX: Governance Mechanisms and Firm Valuations

Read: Table VIII compares ROA, ROE, and net profit margin across high/low governance groups in low-leverage firms. Table IX uses Fama-MacBeth regressions to relate *Q* to internal and external governance proxies and their interactions.

Interpretation: Table VIII supports a real-performance interpretation. Table IX complicates a pure mispricing interpretation: each mechanism alone is valued positively, but the interaction is valued less than additively, which is consistent with higher discount rates or risk in the high-high governance portfolio.

7.6 Table X: Governance Mechanisms and Cash Flow Risk

Table X Governance Mechanisms and Cash Flow Risk The table reports the standard deviations of the four different governance quality groups using accounting measures of return on assets, return on equity, and net profit margin (in percentages). All firms with low leverage (bottom 33% of industry-adjusted leverage) are sorted on their level of takeover vulnerability (external governance) as well as on their level of public pension fund (blockholder) ownership (internal governance), with high and low corresponding to above—and below—median, respectively. The resulting four groups are labeled as High-High, High-Low, Low-High, and Low-Low based on their level of external-internal governance, as measured by EXT, ATI, BLOCK, and PP (see Table II for a description). Net profit margin is the ratio of income before extraordinary items available for common equity to sales; return on equity is the ratio of income before extraordinary items available for common equity to the sum of the book value of common equity and deferred taxes; and return on assets is the ratio of income before extraordinary items to total assets. Each variable is net of the industry median, which is calculated by matching the four-digit SIC codes of all firms in the CRSP-Compustat merged database in December of each year to the 48 industries designated by Fama and French (1997). The average standard deviations for each group are reported, removing outliers (bottom and top 2.5%).			
EXT-INT	ROA	ROE	NPM
Panel A: Different Takeover Vulnerabilities When Internal Governance Is HIGH			
Using PP			
HIGH-HIGH	17.40	12.71	16.13
LOW-HIGH	16.85	11.52	11.97
Using BLOCK			
HIGH-HIGH	16.39	12.35	16.51
LOW-HIGH	14.44	11.06	9.96
Panel B: Different Takeover Vulnerabilities When Internal Governance Is LOW			
Using PP			
HIGH-LOW	16.79	12.77	15.90
LOW-LOW	16.32	12.35	14.10
Using BLOCK			
HIGH-LOW	18.24	13.40	15.55
LOW-LOW	18.27	12.45	15.41

Table X: Governance Mechanisms and Cash Flow Risk

Read: Table X reports the standard deviations of ROA, ROE, and net profit margin for the governance groups.

Economic message: The high-high governance groups tend to have higher cash-flow risk, especially for net profit margin. This is the paper’s strongest evidence that the abnormal returns could reflect a risk premium rather than only market underreaction to governance complementarity.

8 How to read the tables together

8.1 The return evidence

Tables III and IV should be read as a two-way test. Table III asks whether external governance works conditional on the level of internal governance. Table IV asks whether internal governance works conditional on the level of external governance. The answer is symmetric: each mechanism is most important when the other is already strong.

8.2 The interaction evidence

Tables V and VII translate the portfolio findings into a regression structure. *Max* is designed to detect substitution, while *Min* is designed to detect complementarity. The repeated positive *Min* evidence is the statistical implementation of the complementarity claim.

8.3 The fundamentals evidence

Tables VIII through X ask whether abnormal returns correspond to underlying performance or valuation patterns. Profitability is highest when both mechanisms are active, but Q and cash-flow volatility suggest that the high-high portfolio may also be riskier.

9 Mechanism intuition

9.1 Why might blockholders and takeovers complement each other?

A takeover threat is more effective when shareholders can coordinate. A dispersed shareholder base may free ride on takeover gains, making it hard for bidders or activists to discipline management. A large blockholder can reduce coordination frictions, pressure management, or help facilitate value-increasing control changes.

9.2 Why might takeovers make blockholders more effective?

A monitoring shareholder needs an enforcement tool. If management resists change and there is no credible takeover threat, shareholder voice may have limited bite. A contestable control market gives internal monitoring a sharper outside option.

9.3 Why does leverage matter?

Debt may deter takeovers by making acquisition financing harder, reducing free cash flow available to acquirers, or changing bargaining incentives. The paper's low-leverage evidence fits the idea that takeover vulnerability is more valuable when capital structure does not obstruct control changes.

10 Limitations and cautions

- **Not a causal design.** The paper documents robust associations between governance mechanisms and returns/performance. It does not randomly assign governance structures.
- **Governance proxies are imperfect.** *EXT* and *ATI* measure takeover vulnerability through charter and legal provisions, but actual takeover probability also depends on industry, market conditions, size, valuation, and bidder availability.
- **Institutional ownership is endogenous.** Public pension funds and blockholders choose which firms to hold. Their holdings may reflect anticipated performance rather than governance intervention.
- **Return interpretation is ambiguous.** Abnormal returns may reflect mispricing of governance complementarity, omitted risk, or another unobserved factor correlated with the governance portfolios.
- **Time period matters.** The 1990s include major governance changes, takeover activity, and market booms. The authors use intertemporal patterns, but longer samples are needed to fully distinguish learning from risk.

11 Bottom line

The paper's central lesson is that governance mechanisms should not be evaluated one at a time. Takeover vulnerability and active shareholder ownership are most powerful when they coexist. A firm with a contestable control market but no active owner may not be effectively disciplined; a firm with an active owner but strong takeover defenses may also face weak discipline. The empirical implication is that governance research and governance practice should look for systems of mutually reinforcing mechanisms rather than isolated governance scores.